# Md. Abdullah Al Jaber

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## Objective

Aspiring AI/ML Engineer with experience in developing machine learning, deep learning, computer vision, NLP, and federated learning solutions. Passionate about building scalable and privacy-preserving AI systems, with hands-on expertise in TensorFlow, Keras, Scikit-learn, OpenCV, and Flask. Seeking an opportunity to apply analytical and technical skills to solve real-world problems and drive innovation through data-driven technologies.

## Experience

**bKash Limited | Inbound Officer, CC** Nov 2025 – Present

- Handled inbound customer queries with accuracy and professionalism in a fast-paced environment
- Resolved account and transaction issues using internal CRM tools to ensure customer satisfaction
- Collaborated with team members to maintain service quality and response time standards
- Documented customer interactions and escalated complex cases to relevant departments

## Education

**Daffodil International University**
BSc in Computer Science & Engineering **CGPA: 3.30 / 4.00** 2021 – 2025

**Govt. BM College**
Higher Secondary Certificate (HSC) **GPA 5.00 / 5.00** 2018 – 2019

**Jhalakathi NS Kamil Madrasah**
Secondary School Certificate (SSC) **GPA 4.94 / 5.00** 2016 - 2017

## Skills & Abilities

- Machine Learning & AI – N8N Automation, TensorFlow, Keras, PyTorch, Scikit-learn, OpenCV, Pandas, NumPy
- Data Analysis & Visualization – Power BI, Excel, Pandas, NumPy, Matplotlib, Seaborn, Interactive Dashboards, Data Cleaning & Reporting
- Web Development – HTML5, CSS3, JavaScript, React, Next.js, FastAPI, Bootstrap, Flask
- Programming & Tools – Python, C, Java Script, SQL, Git, GitHub, Jupyter Notebook, VS Code, other cloud deployment tools
- Soft Skills – Problem-solving, Communication, Team Collaboration, Continuous Learning

## Projects

- ExpenseFlow – Personal Finance Management Web Application | React, FastAPI, Python, PostgreSQL, JWT
  Developed a full-stack personal finance management application featuring JWT-based authentication, income and expense tracking, transaction management, dashboard analytics, savings insights, and multi-currency support. Integrated a React frontend with a FastAPI backend and PostgreSQL database to provide secure API communication and real-time financial summaries. Tech Stack: React, TypeScript, FastAPI, Python, PostgreSQL (Neon), SQLAlchemy, Alembic, JWT, Vercel, Render.
- Task Manager Web Application | React, FastAPI
  Developed a full-stack task management application featuring JWT authentication, task CRUD, recurring tasks, calendar view, and productivity insights. Tech: React, FastAPI, Python, PostgreSQL/SQLite, JWT.
- AI Resume Matcher Built an AI-powered ATS Resume Analyzer that compares resumes with job descriptions, calculates matching scores, identifies skill gaps, and generates AI-driven career recommendations using Streamlit, LangChain, and Sentence Transformers.
- Bangla News Q&A Bot (RAG) Built a RAG-based system using LangChain and FAISS to process Bangla news articles and answer user questions by retrieving relevant context from news data.
- University PDF Assistant Built an AI-powered PDF Assistant using Python, Streamlit, LangChain, FAISS, Hugging Face Embeddings, and Google Gemini API that enables users to upload PDFs and receive accurate, context-aware answers through Retrieval-Augmented Generation (RAG).

- Fetal Health Prediction (ML Project) Developed machine learning models to predict fetal health conditions based on sensor data. Used Scikit-learn and TensorFlow for classification and model evaluation.
- Image Classification (Computer Vision Project) Implemented a CNN-based image classifier using TensorFlow and Keras for object detection and recognition tasks.
- NLP Sentiment Analysis with Hugging Face And Flask Designed and deployed a text classification model for sentiment analysis using Python and deploy in hugging face and another project is deployed in flask.
- OCR-Based Image-to-Text Extraction (Computer Vision & NLP Project) Developed an OCR system to extract text from scanned images using Tesseract and OpenCV. Processed image data, improved text accuracy through preprocessing, and automated text conversion for document analysis.
- Universal File Converter Developed a web-based file conversion system that allows users to upload and convert multiple file formats into PDF and HTML. Implemented support for DOCX, CSV, XLSX, Python, XML, Jupyter Notebook, and image files with a responsive and user-friendly interface.
- Power BI Sales Performance Dashboard Built an interactive Power BI dashboard to analyze sales performance by location, team, and sales representative. Visualized key metrics including total sales, total boxes sold, and amount per box to support performance monitoring and data-driven decision-making.
- Role-Based Contact Automation | n8n
  Automated contact management with category-based routing, Telegram notifications, Google Sheets logging, and Gmail auto-replies. Tech: n8n, Webhooks, Google Sheets, Gmail, Telegram.
- Enterprise Contact Automation | n8n
  Designed a contact automation workflow with conditional validation, inquiry routing, data logging, and automated notifications. Tech: n8n, Webhooks, JSON, Google Sheets, Gmail, Telegram.

## Research

- Privacy-Preserving Explainable Federated Transfer Learning for Brain Tumor Segmentation
  Contributed to a privacy-preserving federated learning framework for MRI-based brain tumor segmentation, working with CNN-based U-Net and Transformer-based U-Net architectures across centralized, federated, IID, and Non-IID settings. Evaluated model performance using Dice, IoU, Precision, Recall, and F1-score.
- Explainable Hybrid Deep Learning Framework for Bangla Fake News Detection
  Contributed to Bangla fake news detection using BERT and XLM-RoBERTa-based language representations combined with conventional machine learning approaches. Worked with hybrid modeling and explainability techniques, including LIME, to improve model interpretation and analysis.
- Category-Guided Intelligence: Explainable Multi-Task Framework for Bengali Fake News Detection
  Contributed to an explainable multi-task framework for Bengali fake news detection by combining BanglaBERT, BiLSTM, attention mechanisms, and category-aware learning. Worked with LIME, SHAP, and attention-based analysis to interpret model predictions and evaluate model behavior.
- Privacy-Preserving Federated Learning for Pancreatic Malignancy Detection
  Contributed to a federated learning framework for pancreatic cancer detection using SEER clinical data, implementing ANN-based models and FedAvg aggregation in distributed healthcare settings.
- Ensemble Deep Learning for Leukemia Stage Classification
  Contributed to leukemia stage classification using an ensemble of ResNet18 and VGG16 models, with explainable AI techniques for model interpretation.
- Privacy-Preserving Federated Learning Colorectal Cancer Detection
  Developed a privacy-preserving federated learning framework for colorectal cancer detection using distributed healthcare data. Implemented deep learning-based federated training with the FedAvg algorithm and evaluated model performance under distributed learning settings.

## Competitions & Activities

- Achieved 2nd place in a university-wide Data Science contest at Daffodil International University
- Participated in National and University-level Data Science & ML Competitions
- Engaged in Kaggle competitions on data preprocessing, feature engineering, and predictive modeling
- Conducted peer mentoring sessions on Python, ML, and research basics